

# Relation Algebra

## §16. Greedy Algorithms

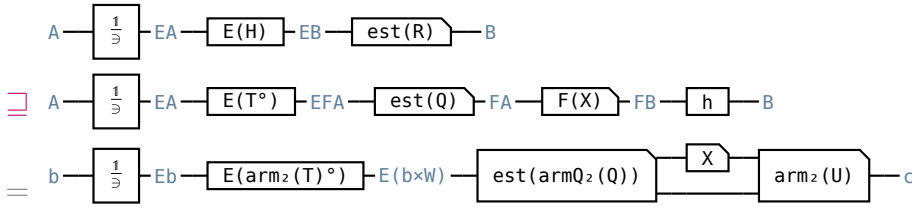
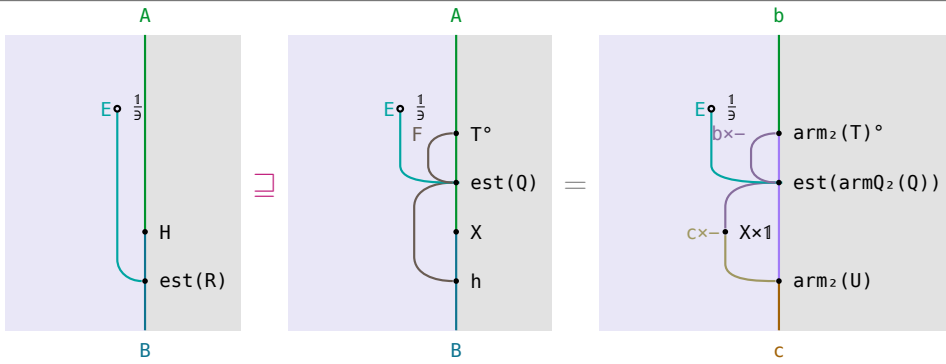
### §16.1. Theory

(16.1a)

$h, T, R, H, M$  as in  $\triangleq H$ ; additionally  $Q$  a **connected** preorder on the sets  $\frac{T^\circ}{\exists}$  returns, so that  $\frac{T^\circ}{\exists} \text{ est}(Q)$  is entire.

$$\text{PreservesRecip}(F) \wedge \text{Map}(h) \wedge \text{MonoAlg}(h, R) \wedge R R \subseteq R \wedge Q F(H) h \subseteq F(H) h R \Rightarrow (\mu X : \frac{T^\circ}{\exists} \text{ est}(Q) F(X) h) \subseteq \frac{H}{\exists} \text{ est}(R) \quad (16.1b)$$

the same optimum reached by keeping ONE decomposition at each step, so that no set is ever carried and the recursion runs on values alone

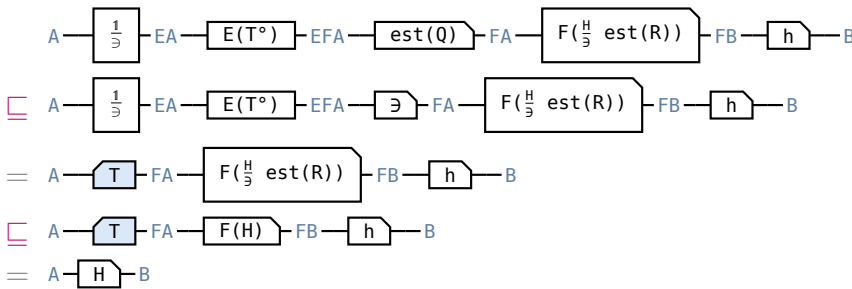
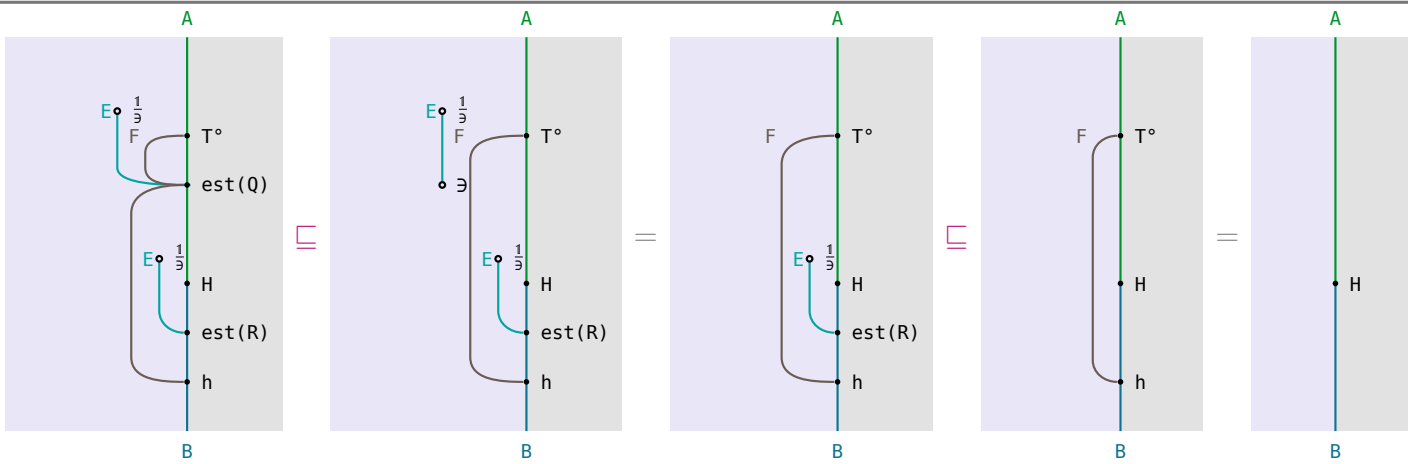


the problem to be solved,  $H \triangleq (T)^\circ (h) \triangleq H, Q$

Proposition 10.1 at  $T=[V_1, V_2]$ ,  $h=[U_1, U_2]$ ,  $Q=Q_1+Q_2$ ,  $V_2 V_1^\circ = \emptyset$

$$T^\circ F(H) h = H \Rightarrow \frac{T^\circ}{\exists} \text{ est}(Q) F(\frac{H}{\exists} \text{ est}(R)) h \subseteq H \quad (16.1c)$$

taking the input apart in one  $Q$ -extreme way, solving the parts by  $M$  and assembling by  $h$  returns only what  $H$  returns



$\text{est}(Q) \subseteq \exists \triangleq \text{est}$

$\frac{T^\circ}{\exists} \exists = T^\circ$  — pow-laws

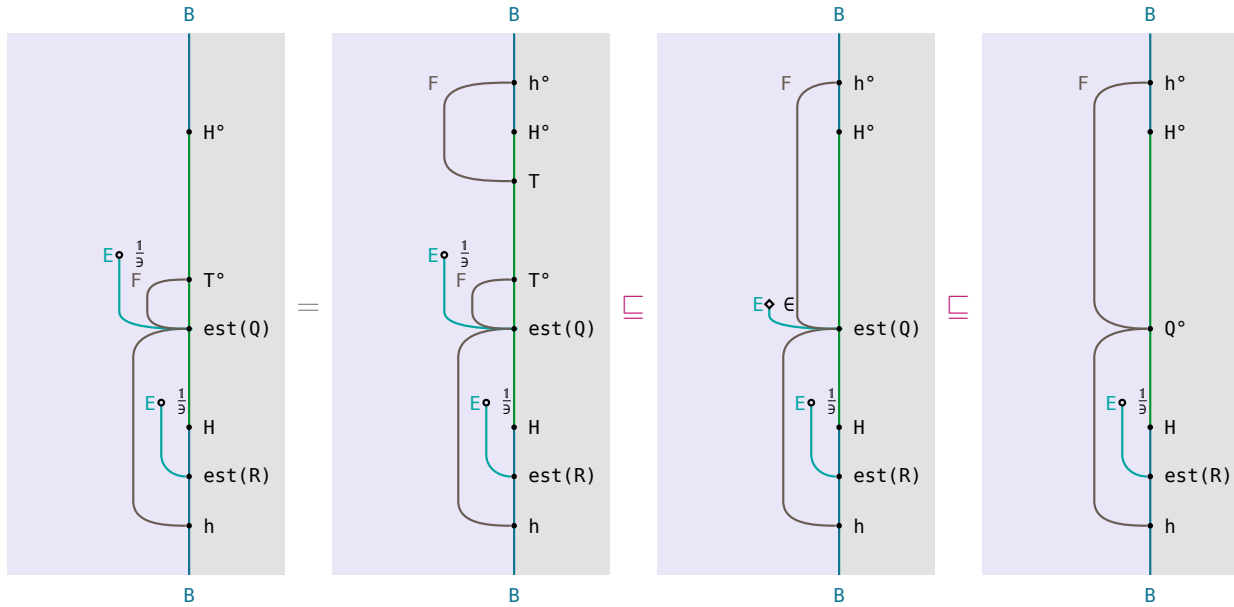
$M \subseteq \frac{H}{\exists} \exists = H$  — est-up

$T^\circ F(H) h = H$ :  $H \triangleq (T)^\circ (h)$  and hylo-fix

$$\text{PreservesRecip}(F) \wedge \text{Map}(h) \wedge \text{MonoAlg}(h, R^\circ) \wedge R^\circ R^\circ \subseteq R^\circ \wedge T^\circ F(H)h=H \wedge QF(H)h \subseteq F(H)hR \Rightarrow H^\circ \frac{T^\circ}{\exists} \quad (16.1d)$$

$$\text{est}(Q)F(\frac{H}{\exists} \text{est}(R))h \subseteq R^\circ$$

$H^\circ$  followed by the greedy body at  $M$  is  $\mathbb{E}R^\circ$ : an answer of the body is never worse than an answer of  $H$  to the same input

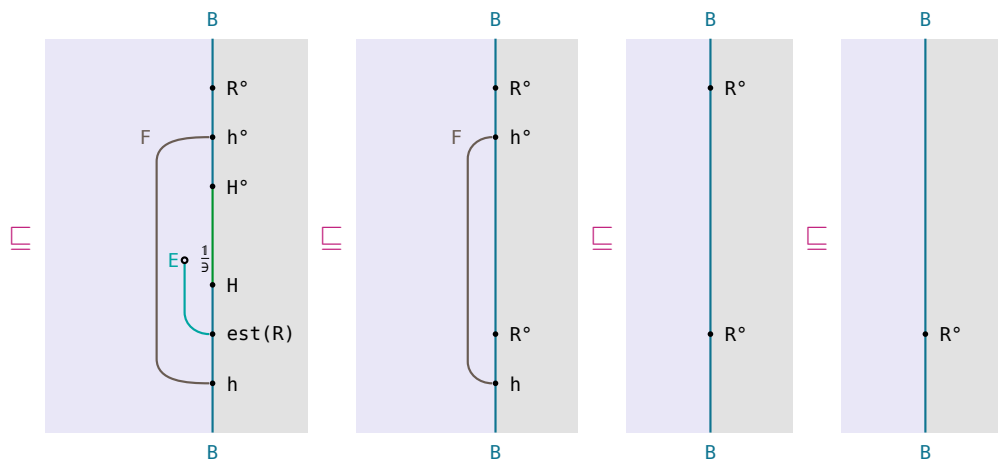


$$\begin{aligned} & B \rightarrow [H] \rightarrow A \rightarrow \left[ \frac{1}{\exists} \right] \rightarrow EA \rightarrow [E(T^\circ)] \rightarrow EFA \rightarrow \text{est}(Q) \rightarrow FA \rightarrow [F(\frac{H}{\exists} \text{est}(R))] \rightarrow FB \rightarrow [h] \rightarrow B \\ & = B \rightarrow [h] \rightarrow FB \rightarrow [F(H^\circ)] \rightarrow FA \rightarrow [T] \rightarrow A \rightarrow \left[ \frac{1}{\exists} \right] \rightarrow EA \rightarrow [E(T^\circ)] \rightarrow EFA \rightarrow \text{est}(Q) \rightarrow FA \rightarrow [F(\frac{H}{\exists} \text{est}(R))] \rightarrow FB \rightarrow [h] \rightarrow B \\ & \sqsubseteq B \rightarrow [h] \rightarrow FB \rightarrow [F(H^\circ)] \rightarrow FA \rightarrow [\epsilon] \rightarrow EFA \rightarrow \text{est}(Q) \rightarrow FA \rightarrow [F(\frac{H}{\exists} \text{est}(R))] \rightarrow FB \rightarrow [h] \rightarrow B \\ & \sqsubseteq B \rightarrow [h] \rightarrow FB \rightarrow [F(H^\circ)] \rightarrow [Q] \rightarrow FA \rightarrow [F(\frac{H}{\exists} \text{est}(R))] \rightarrow FB \rightarrow [h] \rightarrow B \end{aligned}$$

$H^\circ = h^\circ F(H^\circ)T$ , the converse of  $T^\circ F(H)h=H$  — hylo-fix

$T \frac{T^\circ}{\exists} \subseteq \epsilon$ , not a tabulated row:  $\frac{T^\circ}{\exists} \ni T^\circ$  conversed

$\epsilon \text{est}(Q) \subseteq Q^\circ$  — est-up



$$\begin{aligned} & \sqsubseteq B \rightarrow [R] \rightarrow [h] \rightarrow FB \rightarrow [F(H^\circ)] \rightarrow FA \rightarrow [F(\frac{H}{\exists} \text{est}(R))] \rightarrow FB \rightarrow [h] \rightarrow B \\ & \sqsubseteq B \rightarrow [R] \rightarrow [h] \rightarrow [F(R^\circ)] \rightarrow FB \rightarrow [h] \rightarrow B \\ & \sqsubseteq B \rightarrow [R] \rightarrow [R] \rightarrow B \\ & \sqsubseteq B \rightarrow [R] \rightarrow B \end{aligned}$$

$QF(H)h \subseteq F(H)hR$  conversed — the hypothesis on  $Q$

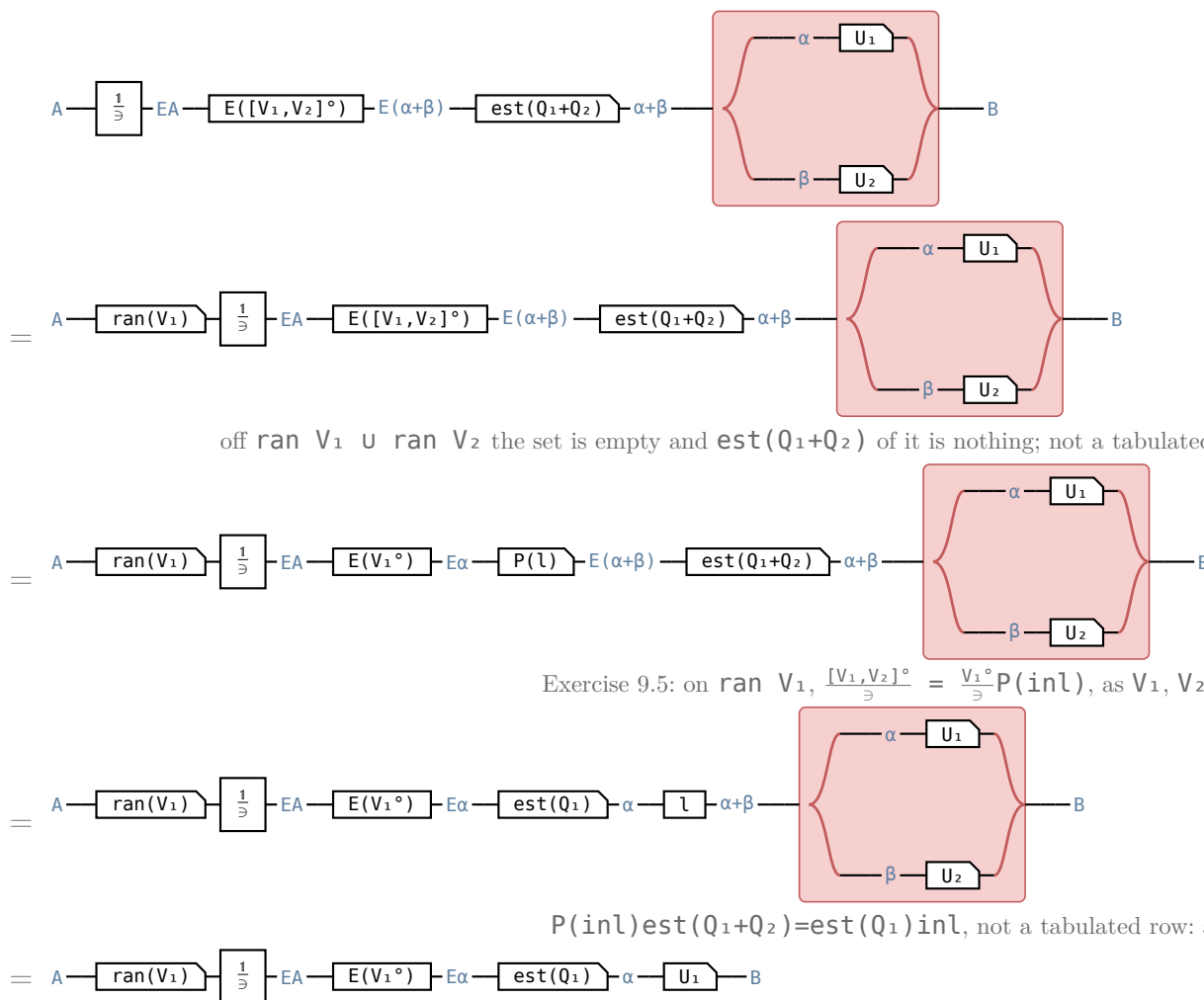
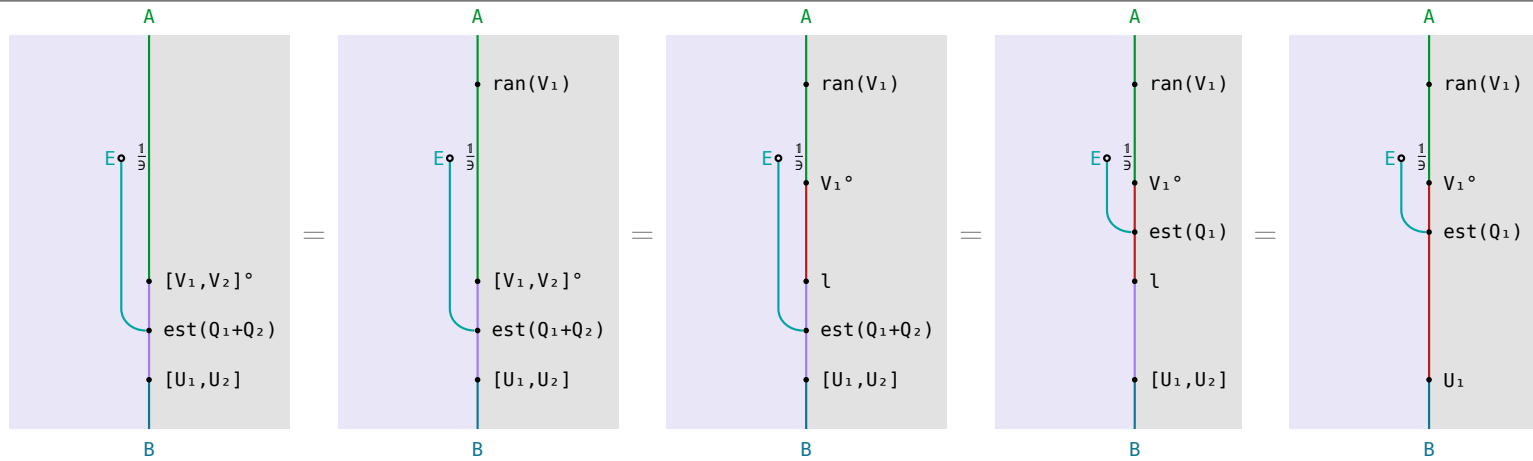
$H^\circ M \subseteq R^\circ$  under  $F$  — est-up

$h^\circ F(R^\circ)h \subseteq R^\circ$ :  $h$  monotonic on  $R$ , shunted — the hypothesis on  $h$

$R$  transitive

$$V_2 \ V_1^\circ = \emptyset \implies \frac{[V_1, V_2]^\circ}{\exists} \text{est}(Q_1+Q_2)[U_1, U_2] = \text{ran}(V_1) \frac{V_1^\circ}{\exists} \text{est}(Q_1)U_1 \cup \text{ran}(V_2) \frac{V_2^\circ}{\exists} \text{est}(Q_2)U_2 \quad (16.1e)$$

when  $V_1$  and  $V_2$  have disjoint ranges, the greedy step over  $[V_1, V_2]$  runs the  $V_1$  step on inputs  $V_1$  reaches and the  $V_2$  step on inputs  $V_2$  reaches



## §16.2. The detab-entab problem

(16.2a)

$\text{detab} \triangleq ([\text{nil}, \text{expand}]) : \text{String} \rightarrow \text{String}$  over  $\text{snoc}$ -lists,  $\alpha \triangleq [\text{nil}, \text{snoc}]$ ,  $H = \text{detab}^\circ$ ;  $\text{expand } (xs, a) = (a = \text{TB} \rightarrow \text{fill } xs, xs \# [a])$ ,  $\text{fill } xs = xs \# \text{blanks } (n - (\text{col } xs) \bmod n)$ .

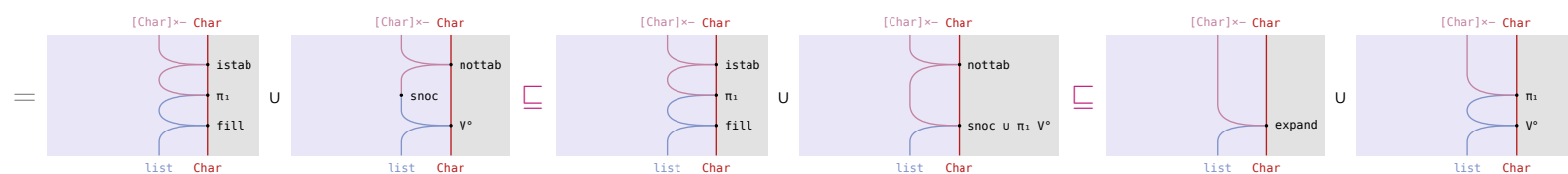
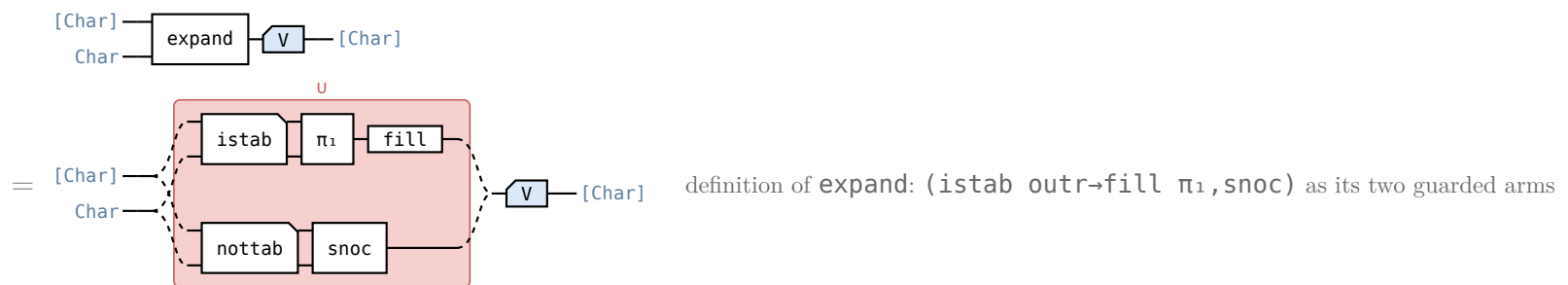
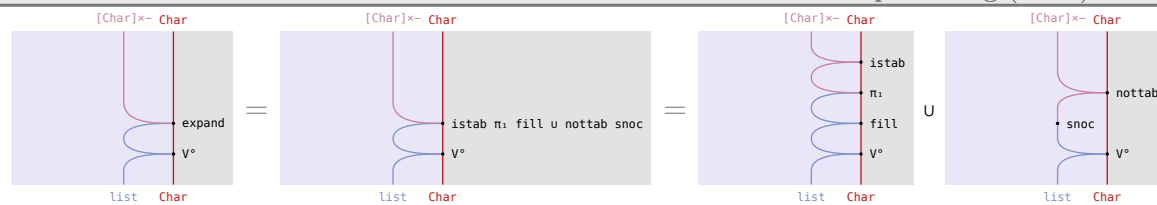
$\text{col} \triangleq ([\text{zero}, \text{count}])$ ,  $\text{count } (c, a) = (a = \text{NL} \rightarrow 0, c+1)$ ; TB the tab, BL the blank, NL the newline, tab stops every  $n$  columns.

$R \triangleq \text{length} \leq \text{length}^\circ$ ,  $U$  the preorder with  $a \cup b \iff a = \text{TB} \vee a = b$ ,  $V \triangleq \text{prefix}^\circ n (\text{fill } \text{fill}^\circ)$ ,  $Q \triangleq 1 + (V \times U)$ .

$\text{unfill } xs$  the shortest prefix of  $xs$  with  $\text{fill}$  ( $\text{unfill } xs = \text{fill } xs$ );  $\text{tbc}$  the trailing blank count,  $\text{triple} \triangleq (\text{unfill } \text{entab}, (\text{tbc}, \text{col}))$ .

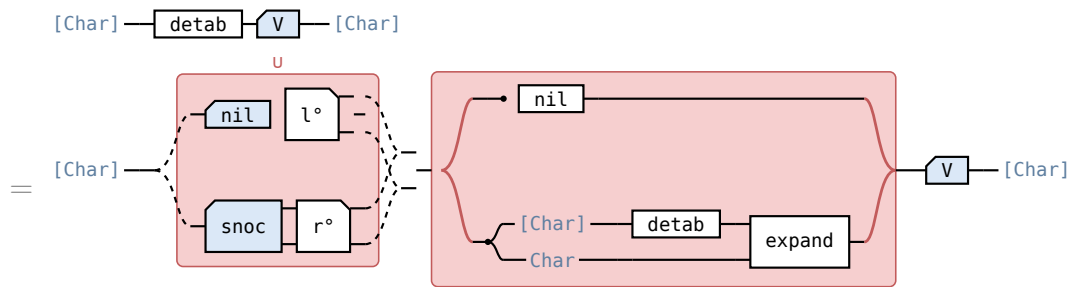
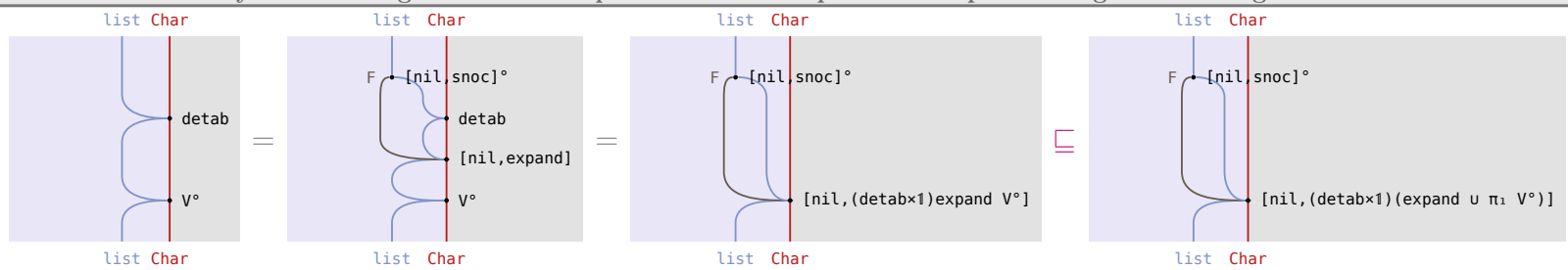
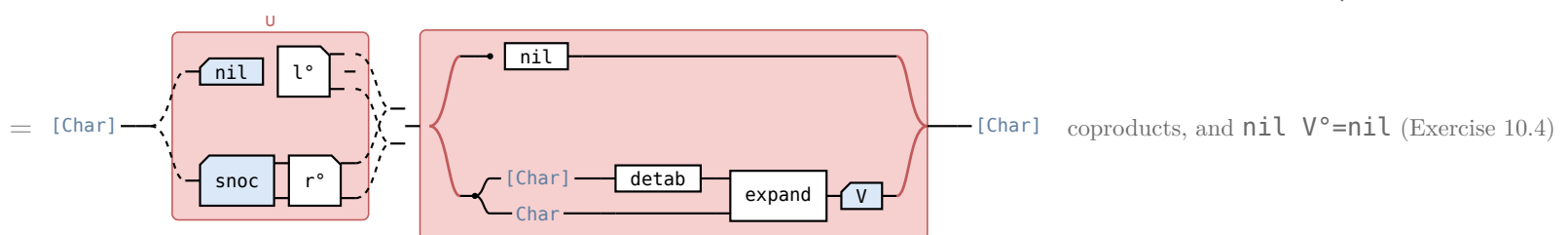
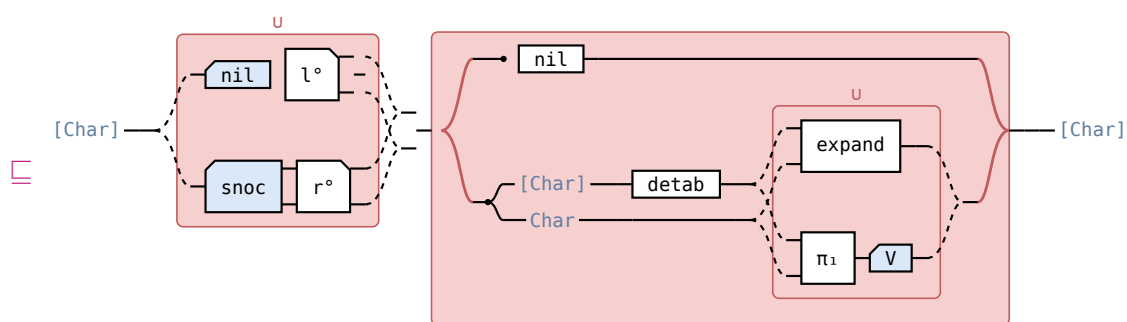
$$0 < n \wedge \text{blank} \neq n1 \Rightarrow \text{expand } V^\circ \sqsubseteq \text{expand} \cup \pi_1 V^\circ \quad (16.2b)$$

shortening the output of one **expand** step to a  $V$ -smaller string either keeps the whole step (**expand**) or drops its character and shortens its input string ( $\pi_1 V^\circ$ )

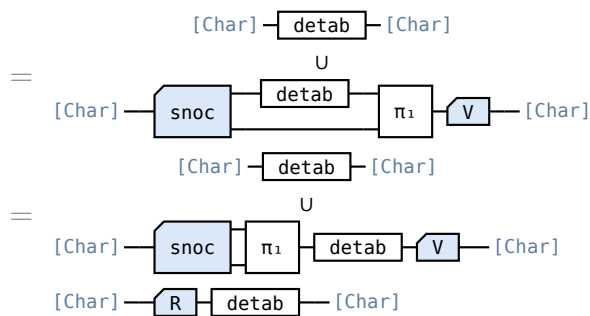
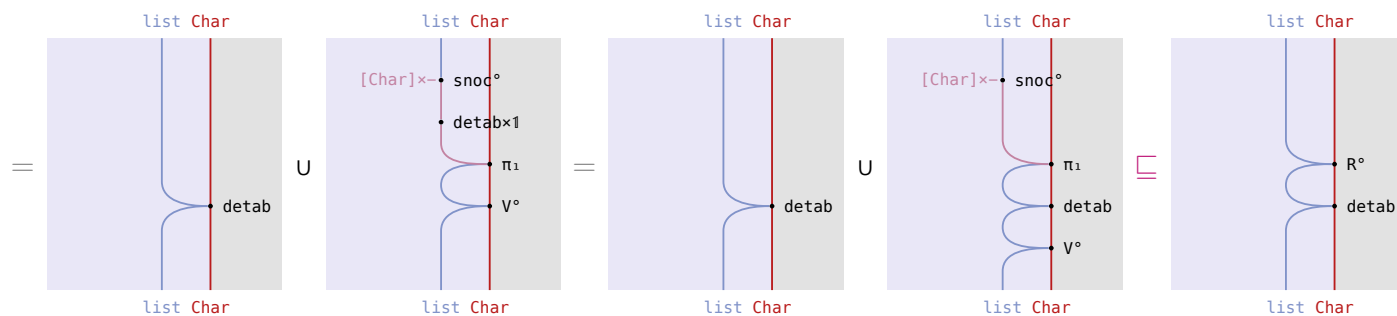


$$0 < n \wedge \text{blank} \neq n1 \Rightarrow \text{detab } V^\circ \sqsubseteq R^\circ \text{detab}$$

(16.2c)

 any  $V$ -shortening of **detab**'s output is **detab**'s output on an input no longer than the given one

**detab** is a fold:  $\text{detab} = \alpha^\circ F(\text{detab}) [\text{nil}, \text{expand}] \triangleq \text{detab}$ 

 coproducts, and  $\text{nil } V^\circ = \text{nil}$  (Exercise 10.4)


the claim — (16.2b)


 distributing  $U$ ; the fold again, and the definition of  $F$ 

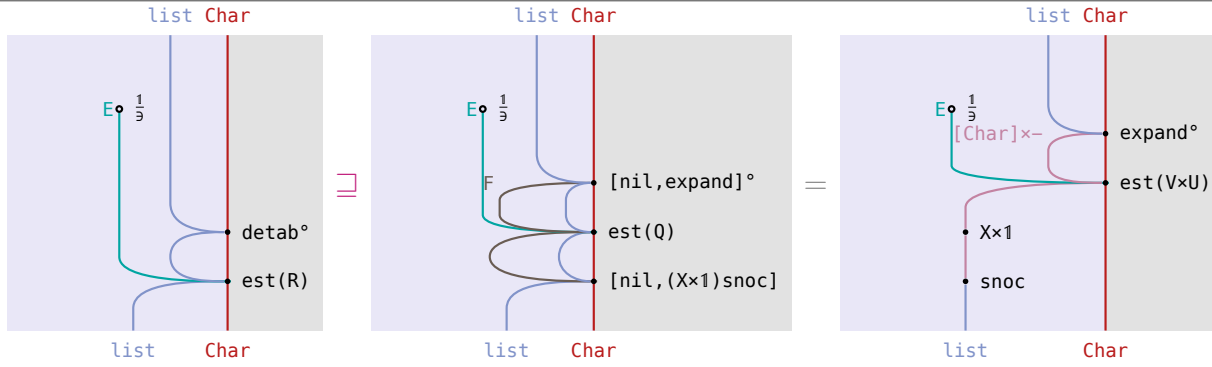
 naturality of  $\pi_1$ :  $(\text{detab} \times 1) \pi_1 = \pi_1 \text{detab}$ ;  $\text{snoc}^\circ \pi_1$  is *init*

$\text{init}$  is inductive, so  $X \triangleq \text{detab } V^\circ$ , a solution of  $X \sqsubseteq \text{detab} u (\text{init } X)$ , lies below the greatest one, **prefix detab** (induction on the input); **prefix**  $\sqsubseteq R^\circ$

$$0 < n \wedge \text{blank} \neq \text{nl} \Rightarrow (\mu X : \frac{[\text{nil}, \text{expand}]^\circ}{3} \text{est}(Q)F(X)[\text{nil}, \text{snoc}]) \sqsubseteq \frac{\text{detab}^\circ}{3} \text{est}(R), \text{ entab} = \text{triple} \quad (16.2d)$$

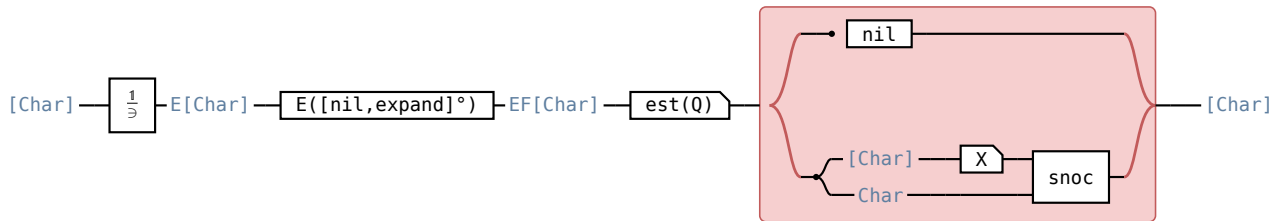
$$\text{assocl } \pi_1 (1 \times \text{blanks}) \text{ cat}$$

the shortest input **detab** expands to the given output is one pass along that output, holding each blank back and cashing the held blanks in for a tab wherever the column reaches a tab stop



$$[\text{Char}] \xrightarrow{\frac{1}{3}} E(\text{detab}^\circ) E[\text{Char}] \xrightarrow{\text{est}(R)} [\text{Char}]$$

the specification  $\triangleq \text{detab}; \text{detab} \text{ entab} = 1$  and nothing shorter does



at  $Q \triangleq 1 + (V \times U)$ : one character of input is decided at each step.  $F(T, R) \alpha \sqsubseteq \alpha R$ . **detab** prefix  $\sqsubseteq R^\circ$  **detab** is FALSE, — at  $n=8$ , **detab**  $[a, b, c, d, e, \text{TB}] = [a, b, c, d, e, \text{BL}, \text{BL}, \text{BL}]$ , whose prefix  $[a, b, c, d, e, \text{BL}, \text{BL}]$  is longer than any input giving it, and **detab**  $V^\circ \sqsubseteq R^\circ$  **detab** holds.  $\text{expand } V^\circ \sqsubseteq \text{expand } U (\pi_1 V^\circ)$  — shortening the output either leaves the last step alone or discards it

$$[\text{Char}] \xrightarrow{\frac{1}{3}} E[\text{Char}] \xrightarrow{E(\text{expand}^\circ)} E([\text{Char}] \times \text{Char}) \xrightarrow{\text{est}(V \times U)} X \xrightarrow{\text{snoc}} [\text{Char}]$$

Proposition 10.1: **nil** and **expand** have disjoint ranges. The greedy step is to emit a tab whenever a tab is legal, consuming all the blanks back to the previous tab stop

$$= \text{entab } xs = \text{entab } (\text{unfill } xs) \# \text{blanks } (\text{tbc } xs) \quad (10.1)$$

what makes  $\text{triple} \triangleq (\text{unfill } \text{entab}, \langle \text{tbc}, \text{col} \rangle)$  a **snoc**-list reduce: the output splits at the last tab stop

$$= \text{triple} = ([\text{base}, \text{op}]), \text{ entab} = \text{triple } \text{assocl } \pi_1 (1 \times \text{blanks}) \text{ cat}$$

base returns  $([], (0, 0))$ ,  $\text{op } ((xs, (t, c)), a) =$

$(a = \text{BL} \wedge (c+1) \bmod n \neq 0 \rightarrow (xs, (t+1, c+1)), a = \text{BL} \rightarrow (xs \# [\text{TB}], (0, c+1)),$

$a = \text{NL} \rightarrow (xs \# \text{blanks } t \# [\text{NL}], (0, 0)), (xs \# \text{blanks } t \# [a], (0, c+1)))$

the program: one pass carrying the column and the count of pending blanks

### §16.3. The minimum tardiness problem

(16.3a)

$\text{FX} = 1 + (X \times \text{Job})$ ,  $\alpha \triangleq [\text{nil}, \text{snoc}]$  on schedules,  $\beta \triangleq [\text{nil}, \text{snag}]$  on bags, **snag** putting a job into a bag;  $\text{bagify} \triangleq \langle \beta \rangle : [\text{Job}] \rightarrow \text{Bag Job}$ ,  $\text{H} = \text{bagify}^\circ$ .

$\text{ct}, \text{dt}, \text{wt} : \text{Job} \rightarrow \text{Real}$  the completion, due and weighting quantities of a job;  $\text{penalty}(xs, j) = (\text{sum}(\text{list}(\text{ct}(xs)) + \text{ct}(j) - \text{dt}(j)) \times \text{wt}(j))$ .

$\text{cost} \triangleq \frac{\text{prefix}}{3} P(\alpha^\circ [\text{zero}, \text{penalty}]) \text{est}(\geq)$ ,  $\text{cost } [] = 0$ ,  $\text{cost } (xs \# [j]) = \text{bmax}(\text{cost } xs, \text{penalty}(xs, j))$ ,  $R \triangleq \text{cost} \leq \text{cost}^\circ$ .

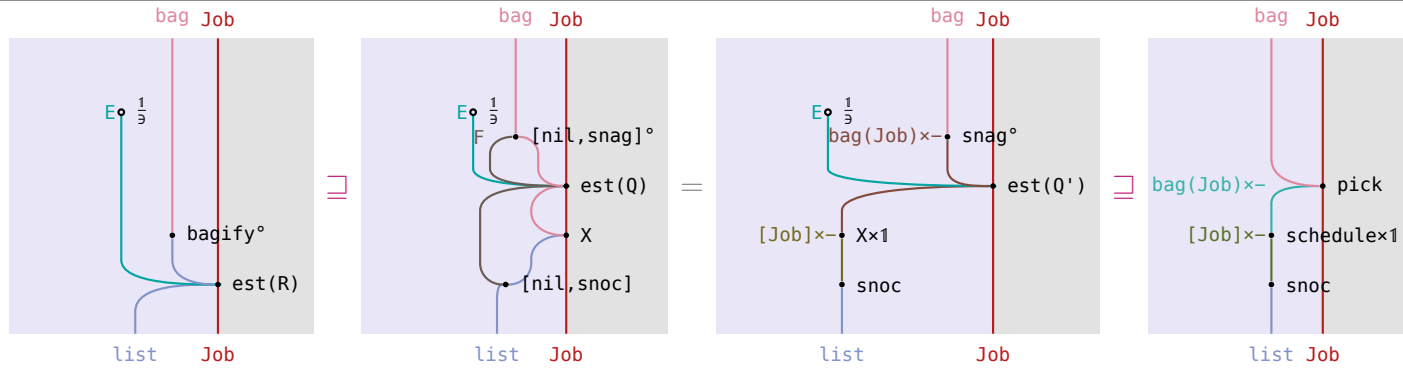
$\text{perm} \triangleq \text{bagify } \text{bagify}^\circ = ([\text{nil}, \text{add}])$ ,  $\text{add } (xs, j) = ys \# [j] \# zs$  for some  $xs = ys \# zs$ .

$k \triangleq [\text{zero}, \text{assocr } (1 \times ((\text{bagify}^\circ \times 1) \text{ penalty})) \text{ bmax}]$ ,  $f \triangleq [\text{zero}, (\text{bagify}^\circ \times 1) \text{ penalty}]$ ,  $Q \triangleq f \leq f^\circ$ ,  $Q' \triangleq (\text{bagify}^\circ \times 1) \text{ penalty} \leq \text{penalty}^\circ (\text{bagify} \times 1)$ .

$$(\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j)) \Rightarrow (\mu X : \frac{[\text{nil}, \text{snag}]^\circ}{3} \text{est}(Q) F(X) [\text{nil}, \text{snoc}] \sqsubseteq \frac{\text{bagify}^\circ}{3} \text{est}(R), \quad (16.3b)$$

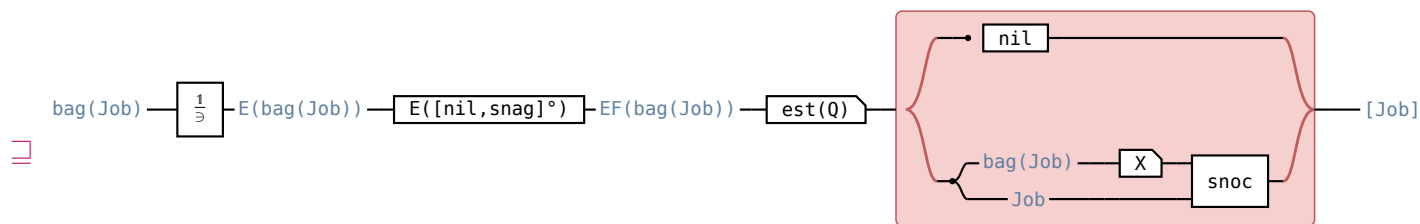
**schedule = (null → nil, pick (schedule × 1) snoc)**

an ordering of the given bag with least maximum penalty is got by taking a job of least penalty out of the bag, putting it last, and scheduling what is left the same way



$$\text{bag}(\text{Job}) \xrightarrow{\frac{1}{3}} E(\text{bag}(\text{Job})) \xrightarrow{E(\text{bagify}^\circ)} E[\text{Job}] \xrightarrow{\text{est}(R)} [\text{Job}]$$

the specification  $\triangleq \text{bagify}$



No greedy **reduce** exists — one would also solve every prefix of the input, and the best schedule of a prefix need not extend to a best schedule of the whole

$$= \text{bag}(\text{Job}) \xrightarrow{\frac{1}{3}} E(\text{bag}(\text{Job})) \xrightarrow{E(\text{snag}^\circ)} E(\text{bag}(\text{Job}) \times \text{Job}) \xrightarrow{\text{est}(Q')} X \xrightarrow{\text{snoc}} [\text{Job}]$$

Proposition 10.1: **nil** and **snag** have disjoint ranges

$$\sqsubseteq \text{bag}(\text{Job}) \xrightarrow{\text{pick}} \text{schedule} \xrightarrow{\text{snoc}} [\text{Job}] \quad \text{pick} \sqsubseteq \frac{\text{snag}^\circ}{3} \text{est}(Q'), \text{ a partial function, quadratic in the number of jobs}$$

## §16.4. The TeX problem

(16.4a)

$\text{intern} \triangleq \text{val round} : \text{Decimal} \rightarrow [0, 2^{16})$ ,  $\text{val} \triangleq ([\text{zero}, \text{shift}])$ ,  $\text{shift}(d, r) = (d + r) / 10$ , round  $r$  rounds  $2^{16}r$  to the nearest integer:  $\text{round } r = n \iff 2n - 1 < 2^{17}r < 2n + 1$ .

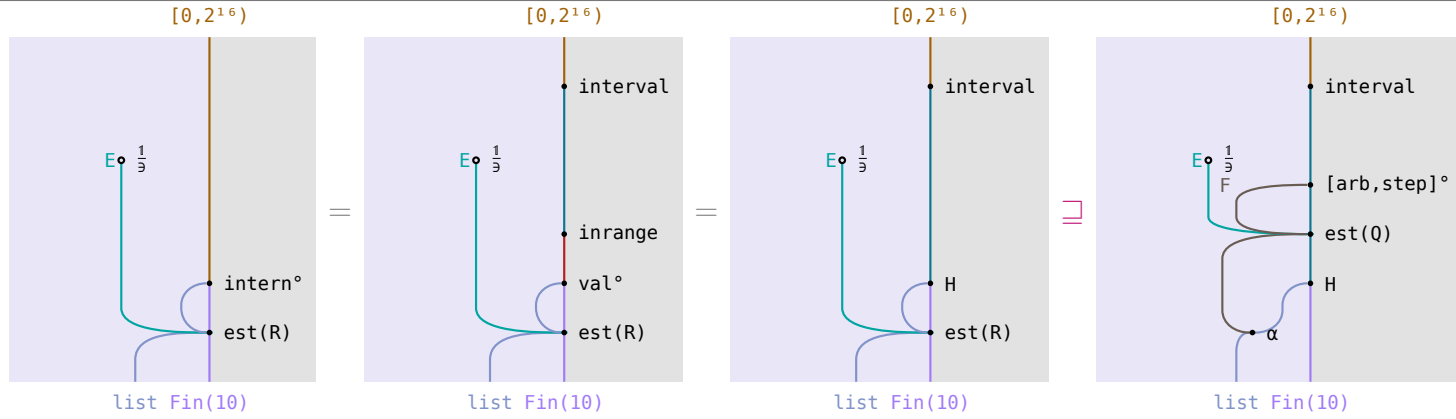
$\text{interval } n = ((2n - 1) / 2^{17}, (2n + 1) / 2^{17})$ ,  $r \text{ inrange } (a, b) \iff a < r < b$ ,  $\text{round}^\circ = \text{interval inrange}$ ,  $R \triangleq \text{length} \leq \text{length}^\circ$ .

Interval all pairs  $(a, b)$ ,  $\text{Legal}(a, b) \iff 0 < b < 1$  and  $a < b$  (10.9), preserved by  $[\text{arb}, \text{step}]$  —  $\text{arb\_legal}$ ,  $\text{step\_legal}$   
 $[\text{arb}, \text{step}] : 1 + (\text{Digit} \times \text{Interval}) \rightarrow \text{Interval}$ ,  $\text{step}(d, (a, b)) = ((d + a) / 10, (d + b) / 10)$ .

$\text{FX} = 1 + (\text{Digit} \times X)$ ,  $\alpha \triangleq [\text{nil}, \text{cons}]$ ,  $H \triangleq ([\text{arb}, \text{step}])^\circ$ ,  $! : \text{Digit} \times \text{Interval} \rightarrow 1$ ,  $Q \triangleq (l^\circ !^\circ r) \cup 1$   $l, r$  are coprod-laws's injections into  $\text{FX} = 1 + (\text{Digit} \times X)$ , so  $l : 1 \rightarrow F(\text{Interval})$  and  $r : \text{Digit} \times \text{Interval} \rightarrow F(\text{Interval})$ ,  $w \triangleq 2^{17}$ .

$$\text{interval}(\mu X : \frac{[\text{arb}, \text{step}]^\circ}{3} \text{est}(Q)F(X)\alpha) \sqsubseteq \frac{\text{intern}^\circ}{3} \text{est}(R) , \quad \text{extern}(n)=f(2n-1, 2n+1) \quad (16.4b)$$

a shortest decimal whose internal representation is the given multiple of  $2^{-16}$  is got by emitting the one digit the interval of admissible reals allows, until that interval contains zero and the empty decimal will do



$$\begin{aligned}
 & [0, 2^{16}) \xrightarrow{\frac{1}{3}} E([0, 2^{16})) \xrightarrow{E(\text{intern}^\circ)} E[\text{Digit}] \xrightarrow{\text{est}(R)} [\text{Digit}] && \text{the specification } \triangleq \text{intern} \\
 = & [0, 2^{16}) \xrightarrow{\text{interval}} \text{Interval} \xrightarrow{\frac{1}{3}} E(\text{Interval}) \xrightarrow{E(\text{inrange val}^\circ)} E[\text{Digit}] \xrightarrow{\text{est}(R)} [\text{Digit}] \\
 & \text{round}^\circ \text{ is not a map, but interval is, so it comes out of the transpose} \\
 = & [0, 2^{16}) \xrightarrow{\text{interval}} \text{Interval} \xrightarrow{\frac{1}{3}} E(\text{Interval}) \xrightarrow{E(H)} E[\text{Digit}] \xrightarrow{\text{est}(R)} [\text{Digit}] \\
 & \text{fusion: val inrange}^\circ = ([\text{arb}, \text{step}])^\circ \text{ — the converse of val, cut down to intervals, is a reduce on cons-lists} \\
 \sqsubseteq & [0, 2^{16}) \xrightarrow{\text{interval}} \text{Interval} \xrightarrow{\frac{1}{3}} E(\text{Interval}) \xrightarrow{E([\text{arb}, \text{step}]^\circ)} EF(\text{Interval}) \xrightarrow{\text{est}(Q)} F(H) \xrightarrow{\alpha} [\text{Digit}] \\
 & \frac{[\text{arb}, \text{step}]^\circ}{3} \text{ returns at most two elements — stop, or take one more digit — and ! nil} \sqsubseteq \text{cons } R^\circ \text{ makes it stop whenever stopping is legal}
 \end{aligned}$$

$$\begin{aligned}
 & = \text{extern} = \text{interval } f, \quad f(a, b) = (a < 0 \rightarrow [], [d] \# f(10a - d, 10b - d)) \\
 & \text{the program, with } d \text{ the digit above}
 \end{aligned}$$

$$\begin{aligned}
 & = \text{extern}(n) = f(2n-1, 2n+1), \quad f(p, q) = (p \leq 0 \rightarrow [], [d] \# f(10p - w \cdot d, 10q - w \cdot d)) \\
 & d = (10q) \text{ div } w: \text{ the same in integer arithmetic only, as chapter 3 required of intern — every interval reached is } (p/w, q/w)
 \end{aligned}$$

## §17. Appendix

### §17.1. $P(S) \text{ est}(R) = (\exists S) n(\epsilon \setminus (SR^\circ))$

(17.1a)

$(p, q)$  tabulates  $W \triangleq (\exists S) n(\epsilon \setminus (SR^\circ))$ , and  $y \triangleq \frac{(p \exists S) n(qR)}{\exists}$ , a map.

(17.1b)

$$\begin{array}{c} \boxed{\begin{array}{c} W \\ P(S) \text{ est}(R) \end{array}} \xleftarrow{p^\circ q = W, 1 \sqsubseteq yy^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \\ \boxed{\begin{array}{c} y^\circ q \\ \text{est}(R) \end{array}} \end{array}$$

(17.1c)

$$\begin{array}{c} \boxed{\begin{array}{c} y^\circ q \\ \exists \end{array}} \xleftarrow{f^\circ \cdot \neg f^\circ, \cdot \exists \neg \exists} \boxed{\begin{array}{c} q \\ (p \exists S) n(qR) \end{array}} \xleftarrow{\Delta \neg n, f^\circ \cdot \neg f^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ q \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} 1 \\ R \end{array}} \end{array}$$

R reflexive

$$\boxed{\begin{array}{c} \epsilon y^\circ q \\ R^\circ \end{array}} \xleftarrow{\cdot \exists \neg \exists, \circ, \text{meet}} \boxed{\begin{array}{c} R^\circ q^\circ q \\ R^\circ \end{array}} \quad \text{q simple}$$

(17.1d)

$$\begin{array}{c} \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \xleftarrow{\Delta \neg n, \cdot T \neg / T, \circ, f^\circ \cdot \neg f^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \end{array}$$

$$\boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \xleftarrow{\cdot \exists \neg \exists, \text{meet}} \boxed{\begin{array}{c} p^\circ p \exists S \\ \exists S \end{array}} \quad \text{p simple}$$

$$\boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \xleftarrow{\text{modular law}} \boxed{\begin{array}{c} p \exists \\ (p \exists) n(qRS^\circ) \end{array}} \xleftarrow{1 \sqsubseteq qq^\circ} \boxed{\begin{array}{c} q^\circ p \exists \\ RS^\circ \end{array}} \xleftarrow{\circ, T \cdot \neg T \setminus} \boxed{\begin{array}{c} W \\ \epsilon \setminus (SR^\circ) \end{array}} \quad \text{W's right half}$$

### §17.2. $P(\text{est}(R)) \text{ est}(R) = P(\text{dom}(\text{est}(R))) \text{ union est}(R)$

(17.2a)

$$\begin{array}{c} \boxed{P(\text{est}(R)) \text{ est}(R)} \xrightarrow{\text{dom}(\text{est}(R)) \text{ est}(R) = \text{est}(R)} \boxed{P(\text{dom}(\text{est}(R)) \text{ est}(R)) \text{ est}(R)} \\ \xrightarrow{P \text{ a relator}} \boxed{P(\text{dom}(\text{est}(R))) P(\text{est}(R)) \text{ est}(R)} \xrightarrow{\text{est-711}} \boxed{P(\text{dom}(\text{est}(R))) \text{ union est}(R)} \end{array}$$

(17.2b)

$$\begin{array}{c} \boxed{\begin{array}{c} P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ P(\text{est}(R)) \text{ est}(R) \end{array}} \xleftarrow{(17.1b) \text{ at } S := \text{est}(R), \Delta \neg n, T \cdot \neg T \setminus} \boxed{\begin{array}{c} P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \exists \text{est}(R) \end{array}} \\ \boxed{\begin{array}{c} \epsilon P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \xleftarrow{\text{count of } T \cdot \neg T \setminus} \boxed{\begin{array}{c} \exists (\exists n(\epsilon \setminus (\epsilon \setminus (SR^\circ)))) \\ \exists (\exists n(\epsilon \setminus R^\circ)) \end{array}} \\ \xleftarrow{\text{modular law}} \boxed{\begin{array}{c} (\exists \exists) n(\epsilon \setminus (\epsilon \setminus R^\circ)) \\ \exists \text{est}(R) \end{array}} \xleftarrow{T(UnV) \sqsubseteq TUnTV, \cdot \exists \neg \exists, \text{est-73}} \boxed{\begin{array}{c} P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \exists \text{est}(R) \end{array}} \\ \xleftarrow{P(\text{dom}(\text{est}(R))) \sqsubseteq 1} \boxed{\begin{array}{c} \text{union est}(R) \\ \exists \text{est}(R) \end{array}} \\ \xleftarrow{\epsilon \text{ lax natural}} \boxed{\begin{array}{c} \epsilon P(\text{dom}(\text{est}(R))) \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \xleftarrow{\text{dom}(\text{est}(R)) \epsilon \text{ union est}(R)} \boxed{\begin{array}{c} \text{dom}(\text{est}(R)) \epsilon \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \\ \xleftarrow{\text{est}(R) \epsilon \text{est}(R)} \boxed{\begin{array}{c} \text{est}(R) \text{ est}(R)^\circ \epsilon \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \xleftarrow{\text{dom}(\text{est}(R)) \sqsubseteq \text{est}(R) \text{ est}(R)^\circ} \boxed{\begin{array}{c} \text{est}(R) \epsilon \text{est}(R) \\ \text{est}(R) R^\circ \end{array}} \quad \text{UP of est}$$

### §17.3. Shortest paths on a cylinder, on lists

(17.3a)

| definition                                                                                                      | type                                               | note                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| $F(A, X) = A + A \times X$                                                                                      | $\mathcal{A} \rightarrow \mathcal{A}$              | The base functor: one square of the new column, alone or in front of the path so far.                                                  |
| $L = \text{list}^+$                                                                                             | $\mathcal{A} \rightarrow \mathcal{A}$              | A path is a non-empty list of squares, one per column crossed.                                                                         |
| $\alpha$ the initial algebra                                                                                    | $F(\text{dCL}(L, E)) \rightarrow \text{dCL}(L, E)$ | $\alpha(5) = [5]$ and $\alpha(1, [5]) = [1, 5]$ : a path is started by one square, or extended by one.                                 |
| $N$<br>the n-tuple relator                                                                                      | $\mathcal{A} \rightarrow \mathcal{A}$              | One component per row of a column, so the fold carries n answers at once.                                                              |
| $R \triangleq \text{sum} \leq \text{sum}^\circ$                                                                 | $[\text{Nat}] \rightarrow [\text{Nat}]$            | The cost of a path, which the cheapest minimises.                                                                                      |
| setify                                                                                                          | $A[n] \rightarrow EA$                              | $\text{setify}(1, 2, 3, 4) = \{1, 2, 3, 4\}$ — which row a component came from is forgotten.                                           |
| moves                                                                                                           | $A[n] \rightarrow E(A[n])$                         | $\text{moves}(x) = \{\text{up}(x), x, \text{down}(x)\}$ — rotated up, unrotated, rotated down.                                         |
| trans                                                                                                           | $E(A[n]) \rightarrow (E A)[n]$                     | $\text{trans}\{(a, b, c), (x, y, z)\} = (\{a, x\}, \{b, y\}, \{c, z\})$ — component k of the result is the set of the k-th components. |
| zip                                                                                                             | $F(a[n]) \rightarrow (F a)[n]$                     | $\text{zip}((1, 2, 3, 4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})) = ((1, \{[5]\}), (2, \{[6]\}), (3, \{[7]\}), (4, \{[8]\}))$ .          |
| $\text{cp} \triangleq \frac{F(1, \exists)}{\exists}$                                                            | $F(EA) \rightarrow EFA$                            | $\text{cp}(1, \{[5], [6], [8]\}) = \{(1, [5]), (1, [6]), (1, [8])\}$ .                                                                 |
| $\text{gen} \triangleq F(1, \text{moves trans } N(\text{union})) \text{zip } N(\text{cp}(F(A, -), T)E(\alpha))$ | $F(NA, N(E(TA))) \rightarrow N(E(TA))$             | $\text{gen}((1, 2, 3, 4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\}))$ is worked out in (17.3.1b).                                           |
| $\text{paths} \triangleq (\text{gen}) \text{setify union}$                                                      | $T(NA) \rightarrow E(TA)$                          | $\text{paths}[(1, 2, 3, 4), (5, 6, 7, 8)]$ is the union of (17.3.1b)'s four sets: 12 paths, 3 from each entry row.                     |
| the specification<br>$\text{paths est}(R)$                                                                      | $[\text{Nat}[n]] \rightarrow [\text{Nat}]$         | A cheapest path from the entry side to the exit side.                                                                                  |

§17.3.1.  $N(E(L A))$ 

$$\begin{array}{l} 1 \quad 5 \\ 2 \quad 6 \\ 3 \quad 7 \\ 4 \quad 8 \end{array} \quad [(1, 2, 3, 4), (5, 6, 7, 8)] : L \ N \ \text{Nat}$$

(17.3.1a)

$(\text{gen})[(5, 6, 7, 8)] = (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})$

(17.3.1b)

$$\begin{aligned} \text{gen}((1, 2, 3, 4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})) = \\ ( \{ [1, 5], [1, 6], [1, 8] \}, \\ \{ [2, 5], [2, 6], [2, 7] \}, \\ \{ [3, 6], [3, 7], [3, 8] \}, \\ \{ [4, 5], [4, 7], [4, 8] \} ) \end{aligned}$$

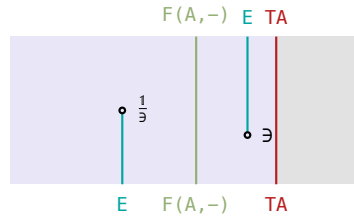
(17.3.1c)

|   |                                                        |
|---|--------------------------------------------------------|
| N | the 4-tuple, one component per row of the entry column |
| E | the set of paths that can start in that row            |
| L | a path is the non-empty list of the squares it crosses |

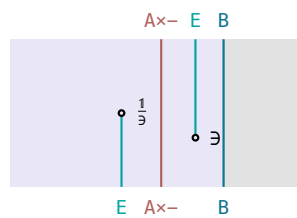
row 1's set has no  $[1,7]$  — from row 1 only rows 4, 1 and 2 are reachable — and it does have  $[1,8]$ , because the cylinder glues the bottom row to the top. That is **moves trans N(union)** of  $\triangleq$  **paths**: component  $k$  collects the paths of rows  $k-1, k, k+1$ .

### §17.3.2. The cross product $cp \triangleq \frac{F(1, \exists)}{\exists}$

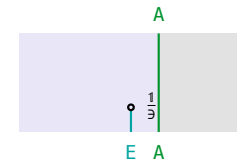
(17.3.2a)



cp on all of  $F(A, E(L A))$



the  $A \times -$  summand:  $\exists$  picks one path,  $1\% \exists$  collects the results



the  $A$  summand, no  $E$  to distribute:  $1\% \exists$  alone,  $a \mapsto \{a\}$

(17.3.2b)

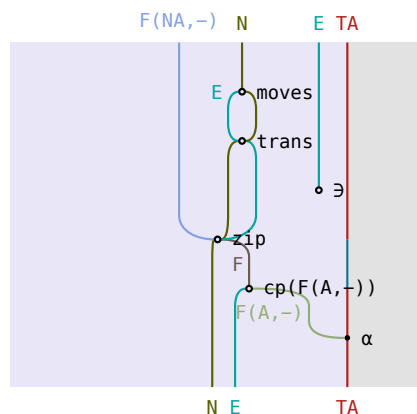
| type                                                         | note                                                                                                   |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| $F(A, -) = A + A \times -$                                   | one wire carries the whole functor, both summands with it; the $\exists$ under it is $F(1, \exists)$   |
| $\exists : E(L A) \rightarrow L A$                           | the $\exists$ inside $F(1, \exists)$ : a set of paths, one of them — $L A$ is one path, $A$ one square |
| $F(1, \exists) = 1 + 1 \times \exists$                       | $: A + A \times E(L A) \rightarrow A + A \times L A$ , the relator acting on each summand              |
| $\exists : E(A + A \times L A) \rightarrow A + A \times L A$ | the $\exists$ under the bar: a set of cells, one of them — a different $\exists$                       |
| $cp : A + A \times E(L A) \rightarrow E(A + A \times L A)$   | $\frac{R}{\exists}$ turns $R : X \rightarrow Y$ into $X \rightarrow E Y$                               |
| $cp = \frac{1}{\exists} E(F(1, \exists))$                    | what the picture draws: the unit makes the outer $E$ , leaving one $\exists$ — pow-laws                |

(17.3.2c)

$w = (1, \{[5], [6], [8]\}) : A \times E(L A)$  the right summand  
 $1 \times \exists : A \times E(L A) \rightarrow A \times L A$   $1$  keeps the square,  $\exists$  picks one path  
 $w \mapsto (1, [5]), (1, [6]), (1, [8]) : A \times L A$  one output per path in  $w$   
 $cp(w) = \{(1, [5]), (1, [6]), (1, [8])\} : E(A + A \times L A)$  the three of them, collected

### §17.3.3. $gen = F(1, \text{moves trans } N(\text{union})) \text{ zip } N(cp P(\alpha))$

(17.3.3a)

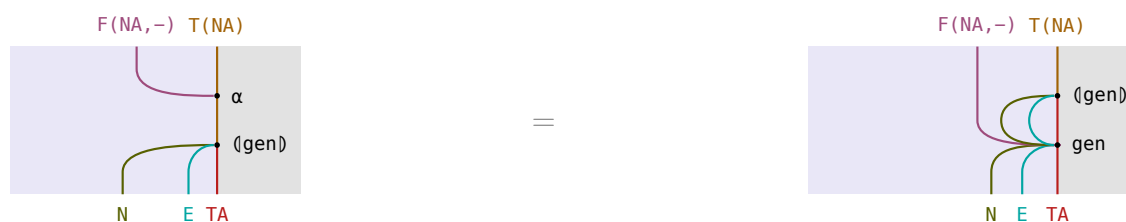


(17.3.3b)

$u = ((1,2,3,4),(\{[5]\},\{[6]\},\{[7]\},\{[8]\}))$  :  $F(N A, N(E(L A)))$   
 $F(1, \text{moves trans } N(\text{union}))$  : 1 keeps the column, the path SETS move  
 $\text{moves}(\{[5]\},\{[6]\},\{[7]\},\{[8]\})$   
 $= \{(\{[6]\},\{[7]\},\{[8]\},\{[5]\}),$   
 $(\{[5]\},\{[6]\},\{[7]\},\{[8]\}),$   
 $(\{[8]\},\{[5]\},\{[6]\},\{[7]\})\}$  :  $E(N(E(L A)))$  down, unmoved, up  
 $\text{trans}(\text{that})$   
 $= (\{[6]\},\{[5]\},\{[8]\},\{[7]\},\{[6]\},\{[5]\}),$   
 $\{[8]\},\{[7]\},\{[6]\},\{[5]\},\{[8]\},\{[7]\})$  :  $N(E(E(L A)))$  row k gets rows k-1, k, k+1  
 $N(\text{union})(\text{that})$   
 $= (\{[5], [6], [8]\},\{[5], [6], [7]\},$   
 $\{[6], [7], [8]\},\{[5], [7], [8]\})$  :  $N(E(L A))$  every path into row k, none dropped  
 $\text{zip}(\text{that})$   
 $= ((1, \{[5], [6], [8]\}), (2, \{[5], [6], [7]\}),$   
 $(3, \{[6], [7], [8]\}), (4, \{[5], [7], [8]\}))$  :  $N(F(A, E(L A)))$  each row: its square, and the paths it may be put in front of  
 $N(\text{cp } P(\alpha))(\text{that})$   
 $= (\{[1,5], [1,6], [1,8]\}, \{[2,5], [2,6], [2,7]\},$   
 $\{[3,6], [3,7], [3,8]\}, \{[4,5], [4,7], [4,8]\})$  :  $N(E(L A))$  cp pairs the square with each path,  $\alpha$  prefixes it:  $\alpha(1, [5]) = [1,5]$

#### §17.3.4. gen is an F-algebra; $(\text{gen}) : \alpha(\text{gen}) = F(1, (\text{gen}))\text{gen}$

(17.3.4a)



$\alpha$  puts the column back on the list, then the fold reads all of it

the fold reads the rest under  $F$ , then one  $\text{gen}$  puts the column in front

(17.3.4b)

$$xs = [(1,2,3,4), (5,6,7,8)] = \alpha((1,2,3,4), [(5,6,7,8)]) : L(N A)$$

$$(\text{gen})(xs) = \text{gen}((1,2,3,4), (\text{gen})[(5,6,7,8)])$$

both halves are in (17.3.1b): the tail folds to  $(\{[5]\},\{[6]\},\{[7]\},\{[8]\})$ , one path per row and each of them one square long, and  $\text{gen}$  on it is (17.3.3b)'s walk.

#### §17.3.5. $Q : F(N A, N(L A)) \rightarrow N(L A)$ , an $F(N A, -)$ -algebra: $Q = F(1, \text{moves trans } N(\text{est}(R))) \text{ zip } N(\alpha)$

(17.3.5a)

$Q : F(N A, N(L A)) \rightarrow N(L A)$  : the fold's algebra: a new column, one path per row  
 $u = ((1,2,3,4),([5],[6],[7],[8]))$  :  $F(N A, N(L A))$   
 $F(1, \text{moves trans } N(\text{est}(R)))$  : 1 keeps the column, the paths move  
 $\text{moves}([5],[6],[7],[8])$   
 $= (\{[6],[7],[8],[5]\},$   
 $\{[5],[6],[7],[8]\},$   
 $\{[8],[5],[6],[7]\})$  :  $E(N(L A))$  down, unmoved, up  
 $\text{trans}(\text{that})$   
 $= (\{[6],[5],[8]\},\{[7],[6],[5]\},$   
 $\{[8],[7],[6]\},\{[5],[8],[7]\})$  :  $N(E(L A))$  row k gets rows k-1, k, k+1  
 $N(\text{est}(R))(\text{that})$   
 $= ([5],[5],[6],[5])$  :  $N(L A)$  the cheapest into each row – chosen BEFORE the new square  
 $\text{zip}(\text{that})$   
 $= ((1,[5]), (2,[5]), (3,[6]), (4,[5]))$  :  $N(F(A, L A))$  each row: its square, and the one predecessor that survived  
 $N(\alpha)(\text{that})$   
 $= ([1,5], [2,5], [3,6], [4,5])$  :  $N(L A)$   $\alpha(1, [5]) = [1,5]$

| $Q = [N(\text{wrap}), (1 \times \text{moves trans } N(\text{est}(R))) \text{ zip}' N(\text{cons})]$ (17.3.5b)                                                                                                                                                                                                                                                  |                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| at the last column $Q$ starts one path per row, and at each earlier one it puts each square in front of the cheapest of the three kept paths it can step to — the algebra (17.3.5c)'s last step folds                                                                                                                                                          |                      |
| <b>circuit</b> — the fork is $F(NA, N(LA)) = NA + NA \times N(LA)$                                                                                                                                                                                                                                                                                             | <b>Hinze–Marsden</b> |
| $F(N A) (N (T A)) \text{---} \boxed{Q} \text{---} N(T A)$                                                                                                                                                                                                                                                                                                      |                      |
| $FN(T A) \text{---} \boxed{F(1, \text{moves trans } N(\text{est}(R)))} \text{---} FN(T A) \text{---} \boxed{\text{zip}} \text{---} N(F A (T A)) \text{---} \boxed{N(\alpha)} \text{---} N(T A)$<br>the fusion condition (17.3.5d) read as a definition,                                                                                                        |                      |
| $FN(N A) (N (T A)) \text{---} \left\{ \begin{array}{l} N(p_{\circ}) \text{---} \boxed{N(\text{wrapA})} \\ q \text{---} \boxed{\text{pm}} \text{---} q \text{---} \boxed{\text{zip}'} \text{---} N(q_{\circ}) \text{---} \boxed{N(\text{consA})} \end{array} \right\} \text{---} N(T A)$<br>$\text{zip} = 1 + \text{zip}', \alpha = [\text{wrap}, \text{cons}]$ |                      |

| <b>paths <math>\text{est}(R) \exists (Q)</math> setify <math>\text{est}(R)</math></b> (17.3.5c)                                                                                                                                                                                                                                                                                                                                               |                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| shortest paths on a cylinder: the dynamic programming approach to exhaustive search allows a path of least cost to be found in $O(n \times m)$ time; $Q$ is (17.3.5b)'s algebra.                                                                                                                                                                                                                                                              |                      |
| <b>circuit</b>                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Hinze–Marsden</b> |
| $T(N\ A) \text{---} \boxed{\text{paths}} \text{---} E(T\ A) \text{---} \boxed{\text{est}(R)} \text{---} T\ A$                                                                                                                                                                                                                                                                                                                                 |                      |
| $= T(N\ A) \text{---} \boxed{F(N\ A)\ (N\ (E\ (T\ A)))} \text{---} \boxed{\text{gen}} \text{---} NE(T\ A) \text{---} \boxed{\text{setify}} \text{---} EE(T\ A) \text{---} \boxed{\text{union}} \text{---} E(T\ A) \text{---} \boxed{\text{est}(R)} \text{---} T\ A$ $\triangleq \text{paths at paths}$                                                                                                                                        |                      |
| $\sqsubseteq T(N\ A) \text{---} \boxed{F(N\ A)\ (N\ (E\ (T\ A)))} \text{---} \boxed{\text{gen}} \text{---} NE(T\ A) \text{---} \boxed{\text{setify}} \text{---} EE(T\ A) \text{---} \boxed{P(\text{est}(R))} \text{---} E(T\ A) \text{---} \boxed{\text{est}(R)} \text{---} T\ A$ $P(\text{est}(R))\ \text{est}(R) \sqsubseteq \text{union}\ \text{est}(R) \text{ --- a minimum in each set, then a minimum of those; } R \text{ transitive}$ |                      |
| $\sqsubseteq T(N\ A) \text{---} \boxed{F(N\ A)\ (N\ (E\ (T\ A)))} \text{---} \boxed{\text{gen}} \text{---} NE(T\ A) \text{---} \boxed{N(\text{est}(R))} \text{---} N(T\ A) \text{---} \boxed{\text{setify}} \text{---} E(T\ A) \text{---} \boxed{\text{est}(R)} \text{---} T\ A$ $\text{setify lax natural}$                                                                                                                                  |                      |
| $\sqsubseteq T(N\ A) \text{---} \boxed{F(N\ A)\ (N\ (T\ A))} \text{---} \boxed{Q} \text{---} N(T\ A) \text{---} \boxed{\text{setify}} \text{---} E(T\ A) \text{---} \boxed{\text{est}(R)} \text{---} T\ A$ $\text{fusion at (17.3.5d)}$                                                                                                                                                                                                       |                      |

| <b>gen <math>N(\text{est}(R)) \exists F(1, N(\text{est}(R))) Q</math></b> (17.3.5d)                                                                                                |                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| fusion: the condition for fusion in (17.3.5c)'s last step, used to derive a definition of $Q$ .                                                                                    |                      |
| <b>circuit</b>                                                                                                                                                                     | <b>Hinze–Marsden</b> |
| $F(N\ A)\ (N\ (E\ (T\ A))) \text{---} \boxed{\text{gen}} \text{---} NE(T\ A) \text{---} \boxed{N(\text{est}(R))} \text{---} N(T\ A)$                                               |                      |
| $\sqsubseteq FNE(T\ A) \text{---} \boxed{F(1, N(\text{est}(R)))} \text{---} FN(T\ A) \text{---} \boxed{Q} \text{---} N(T\ A)$ $(7.13), \text{ then zip, trans, moves lax natural}$ |                      |

(7.13) is  $F(1, \text{est}(R)) \alpha \sqsubseteq \text{cp } P(\alpha)\ \text{est}(R)$ , mon-thm71 at the map  $\alpha$  with  $\frac{F(1, \exists)\alpha}{\exists} = \text{cp } P(\alpha)$ : extending every path in a set and then taking a minimum is beaten by extending one minimum. It is the crux here, not the greedy theorem.